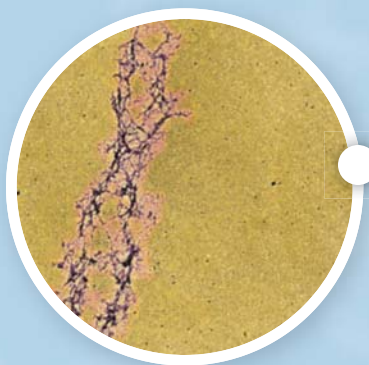


Summary



ESSENTIAL QUESTION

How is DNA the foundation for unity and diversity of living things?



TOPIC 1.1:

How does an understanding of DNA help us investigate living things?

- The variation in living things we see around us is due to DNA.
- DNA is made of many nucleotides linked together in a specific order.
- DNA exists in chromosomes, which contain thousands of genes.
- The structure of DNA is important to passing on genetic information.
- The different genetic make-up of organisms is reflected in the diversity of life.

Key Terms

DNA	nucleotide
protein	chromatin
chromosome	gene
allele	karyotype
species	population
nitrogenous bases	
complementary bases	
homologous chromosome	

TOPIC 1.2:

How is hereditary information passed from one generation to the next?

- Genes pass on inherited traits from parent to offspring.
- Punnett squares show the probability of offspring inheriting specific traits.
- Both alleles are expressed in codominance.
- In incomplete dominance, alleles are neither dominant nor recessive.
- Some inherited traits are due to alleles on the sex chromosomes.

Key Terms

genetics	traits	dominant
recessive	phenotype	genotype
homozygous	heterozygous	codominance
incomplete dominance		sex-linked traits



TOPIC 1.3:

How can natural and artificial selection influence changes in populations?

- DNA mutations produce genetic diversity within a population.
- Natural selection favours traits that make an organism better suited to its environment.
- Natural selection can lead to the formation of new species.
- Environmental factors can cause mutations.
- Humans select desired characteristics in organisms to be passed on to the next generation.

Key Terms

mutation	selective advantage	natural selection
adaptation	adaptive radiation	extinction
mutagen	carcinogen	artificial selection
monoculture		



TOPIC 1.4:

How and why are the genes of organisms manipulated?

- DNA of a living cell can be copied, modified, and inserted into another organism.
- DNA technology has many uses.
- The use of biotechnology has some risks and raises some ethical issues.

Key Terms

biotechnology	cloning
gene cloning	recombinant DNA
artificial insemination	in vitro fertilization (IVF)
gene therapy	

Review

What Do You Know? Connecting to Concepts

Visualizing Ideas

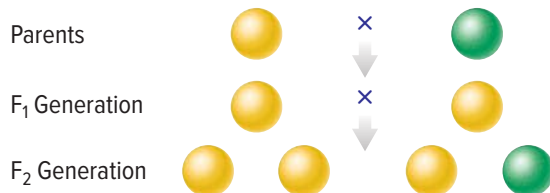
1. Study the photo here. How does DNA account for both the similarities and differences among the different varieties of apples?



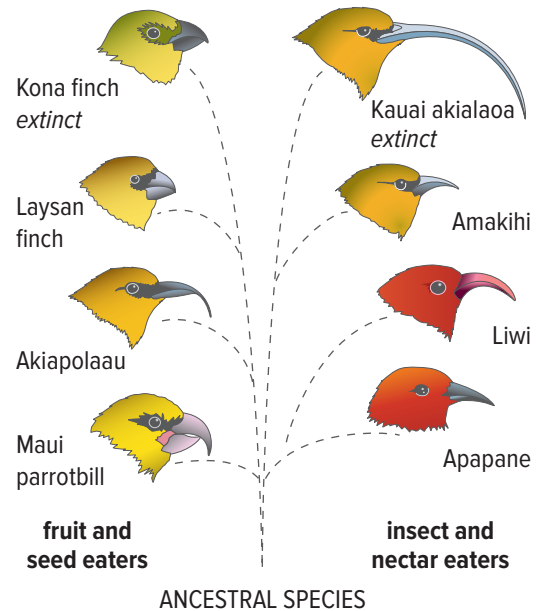
2. Suppose you have to explain the concept of homologous chromosomes to a Grade 6 class. How could you use the diagram on the right as part of your explanation? What labels would you add?



3. The diagram below shows the results of two crosses. Explain the results and the genetic principle that is illustrated.



4. The image at the top of the next column shows what happened over time to a group of birds called honeycreepers after an ancestral species reached the Hawaiian Islands. Each island has different biotic and abiotic conditions. Identify and describe the concepts shown.



Using Key Terms

5. Create a table with three columns. In the first column, list all the key terms from this unit. In the second column, record a definition for each term, written in your own words. In the third column, sketch or draw a small picture that can help you remember the term and its meaning.
6. In a format of your choice, show how the term *DNA* relates to following terms from this unit:
 - gene
 - species
 - population
 - traits
 - mutation
 - natural selection
 - adaptation
 - adaptive radiation
 - artificial selection
 - monoculture
 - biotechnology

Communicating Concepts

7. Scientists often describe the structure of DNA by comparing it to a ladder. Draw a DNA molecule and use the drawing to explain how DNA is similar to a ladder. Then explain the limitations of this comparison.

8. DNA sequences in a genome are compared to letters strung together in a book. Develop another analogy for how DNA, chromosomes, genes, and nucleotides are related.
9. Sketch a small section of DNA. Use the sketch to describe how DNA's complementary nature enables accurate replication.
10. "The sex chromosomes in a human are a homologous pair." Do you agree or disagree with this statement? Explain why.
11. Distinguish between the following pairs of terms using a definition and an example.
 - a) homozygous and heterozygous
 - b) dominant and recessive
12. Given the phenotype of an individual, can you determine its genotype? Explain.
13. Since Mendel performed his experiments with pea plants, scientists have discovered that there are more complex patterns of inheritance. Use examples and diagrams to illustrate the differences among the following mechanisms:
 - dominance
 - incomplete dominance
 - codominance
 - sex-linked inheritance
14. In a Venn diagram, compare and contrast natural selection and artificial selection.
15. Evolution is the process of biological change over time based on the relationships between species and their environments. Summarize how genetic variation is involved in biological change over time.
16. Make a table to show three examples of transgenic organisms and describe their practical application.
17. Make a flowchart to show the steps involved in gene cloning.
18. There are many benefits to biotechnology, but there are also ethical concerns. Use a concept map to illustrate some of the benefits and concerns that are associated with the biotechnology topics discussed in this unit.

What Do You Know?

Connecting to Competencies

Developing Skills

19. A female fruit fly that is homozygous dominant for red eyes is crossed with a white-eyed male fruit fly. Use a Punnett square to predict the genotype(s) and phenotype(s) of their offspring.
20. Explain why there are varieties of houseflies that are not killed by a popular insecticide. Under what conditions is such resistance considered to be a selective advantage?
21. Outline a breeding program that would help you develop a cow that produces more milk.
22. Choose a species of animal, and describe two of its traits. Explain the adaptive value of both traits. Include a sketch with your explanation.
23. Research a drug or other form of medical treatment that was developed using recombinant DNA technology. Describe what it is, its use, and any risks or controversies that are associated with its use.

Thinking Critically and Creatively

24. Describe an experimental approach to determine which of two alleles for a gene is recessive.
25. Develop a plot for a movie or play that involves the use of gene therapy. Ensure that your ideas are scientifically plausible.

Unit 1 Review *(continued)*

26. In radishes, colour is controlled by two alleles, one for red colour and one for white colour. Inheritance of these alleles shows incomplete dominance. The photographs below show the phenotype for each possible colour: red, purple, and white. What phenotypic ratio would you expect from crossing two heterozygous radish plants?



27. State whether each of the following examples demonstrates artificial selection. Explain why or why not.
- a) A person breeds long-furred Persian cats.
 - b) A farmer increases the variability in the fat content of different plant species.
 - c) Over time, plants develop chemical defences that deter herbivores.
 - d) Crops are bred to be pest-resistant.

Understanding Big Ideas Making New Connections

28. Imagine that you are a journalist writing an article for the magazine *Ethics in a Changing World*. You have been asked to research and explain the social and environmental implications of current genetics research. Suggested topics include gene cloning, transgenic crops to reduce hunger, in vitro fertilization, and gene therapy. Choose one topic and write an article on what you find.

Applying Your Understanding

29. A black-haired true-breeding guinea pig is crossed with a white-haired true-breeding guinea pig. All the offspring have black hair.
- a) Which hair colour is dominant?
 - b) What are the genotypes and phenotypes of the parents?
 - c) What are the genotypes and phenotypes of the offspring?
30. A yellow-haired rat is mated with a black-haired rat. Over time, the rats produce 45 black offspring and 52 yellow offspring. From these results, determine the *probable* genotypes of the parents and offspring. How could you determine the dominant allele?
31. Sickle cell anemia is an autosomal recessive disorder that results in the formation of abnormally shaped red blood cells. Write the genotypes for the following individuals:
- a) a person with sickle cell anemia
 - b) a person carrying the sickle cell allele
 - c) a homozygous person with a normal phenotype
32. Hemophilia is a disease that is due to an X-linked recessive allele. A man and woman who do not have hemophilia have a son who does have hemophilia. Explain how this can occur. If they have a daughter, could she have hemophilia? Explain.

Thinking Critically and Creatively

33. Sea stars eat clams by pulling apart the two halves of a clam's shell. Create a sequence of drawings to show how this could result in natural selection of muscle size in clams.
34. Adaptive radiation often occurs on islands. Would you expect to find more adaptive radiation on islands that are remote from the mainland, like small Pacific islands, or islands that are close to the mainland, like Vancouver Island? Explain your answer.

35. Explain the following observations from a scientific research study that involved finches (*Geospiza fortis*) on the Galapagos Islands.

- During a drought in 1977, a large percentage of *Geospiza fortis* finches died of starvation on the Galapagos island Daphne Major.
- The 90 bird survivors in 1978 had a beak depth from about 9.4 mm to about 10.2 mm, which was greater than the beak depth of the finches that did not survive.

36. Scientists at the University of Guelph produced genetically modified pigs. These pigs were given the name Enviropig™ because they were genetically altered so that less phosphorus was excreted in their waste. High levels of phosphorus in animal waste can enter streams and lakes and act as a pollutant.

- a) What is one advantage of developing these pigs?
- b) What is a potential risk of these pigs being produced?
- c) These pigs were being considered as a source of food. Should genetically modified food that is to be consumed by people be labelled as such? Provide a supporting statement for your opinion.

Connect to Self and Society

37. In Canada, individual grizzly bears and populations of grizzly bears are being isolated as humans expand their use of land that was previously used by the bears.

- a) If the grizzly bear were to become extinct, what might some of the economic, political, and social implications be for Canada?
- b) How might wildlife corridors help the situation? (Wildlife corridors are routes designed to help animals cross busy highways safely in mountain parks.)

38. Indoor tanning beds work by exposing people to artificial ultraviolet light. Researcher have found that this increases a person's risk of developing a deadly form of skin cancer, called melanoma, by 20%.

- a) What do you know about the link between ultraviolet light and cancer?
- b) Several countries have passed laws to restrict the use of tanning beds by teenagers. What are the regulations in British Columbia for the use of indoor tanning beds? Do you think these regulations are strict enough, too strict, or not strict enough? Why?

39. Write a paragraph that summarizes how you think society has been affected by biotechnology and what the impact has been to date. In your paragraph be sure to include examples and information that support your opinion.

40. Choose a crop plant that is grown in British Columbia that has been genetically modified to be more resistant to disease. Research the economic and environmental impacts that growing this crop has had in B.C. Give your opinion as to whether the crop should continue to be grown in B.C. or not. Use evidence from your research to support your opinion.

41. With modern technologies, scientists can retrieve DNA from 5000 year old bones found in archeological digs. Recently some First Nations people have shared their DNA with scientists to see if there are connections with the DNA of people who lived in the same area thousands of years before. The results of these studies often show direct links between contemporary First Nations and their ancestors. In what ways does this research help to bring together Indigenous scientific knowledge and Western science?



Unit Assessment

How can we use our knowledge of DNA to learn more about the past and the future?



Black-footed ferrets were once plentiful in the prairies of North America. However, disease and the decline of their main food source left only 18 individuals by the mid-1980s. To save the species from extinction, they were captured as part of a zoo program using selective breeding and artificial insemination to reintroduce genetic diversity into the population. In 2009, 34 ferrets were released back to the wild—the program having been

a success. The burrowing owl is another example of a species in which wild populations were increased due to a breeding program at the BC Wildlife Park (BCWP). After being locally extinct from British Columbia by 1980, the BCWP began a breeding program and, since 1991, over 2 500 burrowing owls have been released back into the wild. The BCWP is still working towards increasing the genetic diversity of local populations.

Work as part of a group to do the following.

- STEP 1** ▶ Reflect on the three options, their photos, and the question asked for each option.
- STEP 2** ▶ Brainstorm three more options and questions of your own about a situation that involves DNA and how our knowledge can be used to solve a problem or broaden our thinking.
- STEP 3** ▶ Decide on one of the six options questions to investigate.
- STEP 4** ▶ Plan and conduct a scientific inquiry to explore your question.
- STEP 5** ▶ Organize and analyze the data and information that you find and collect.
- STEP 6** ▶ Communicate the results of your inquiry in a suitable manner.

OPTION A

Evolutionary Relationships

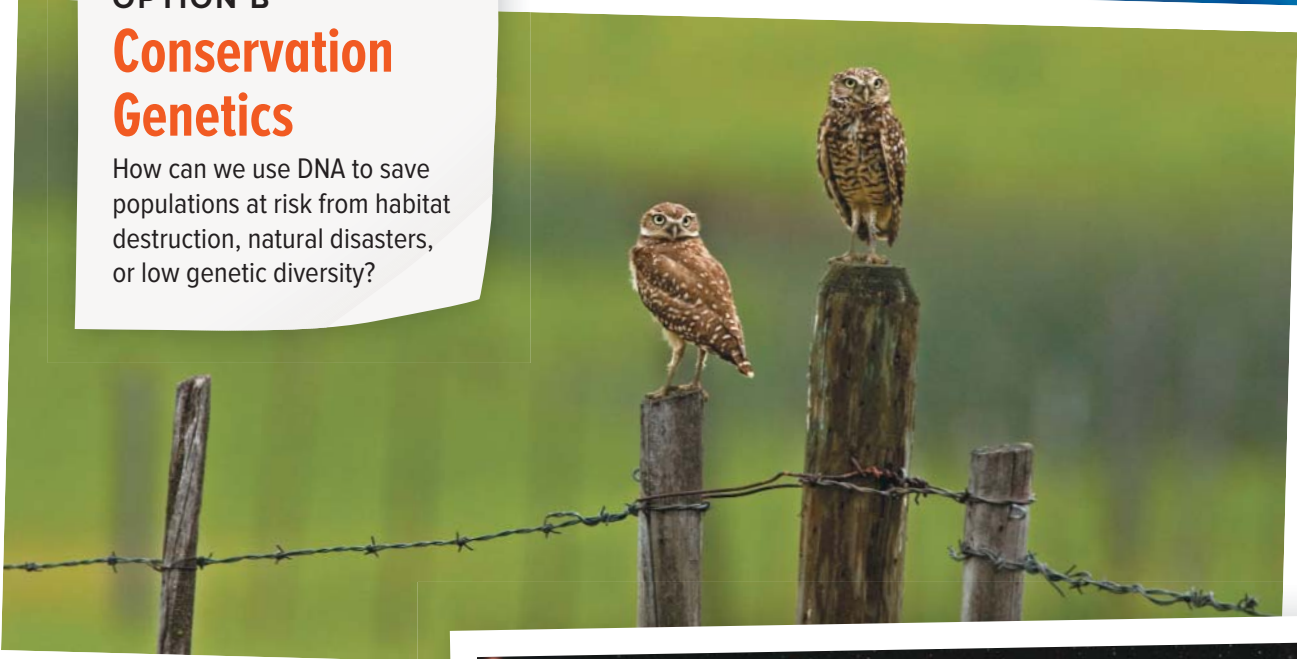
What can we learn about evolutionary relationships among organisms by comparing DNA from fossils to that of modern organisms?



OPTION B

Conservation Genetics

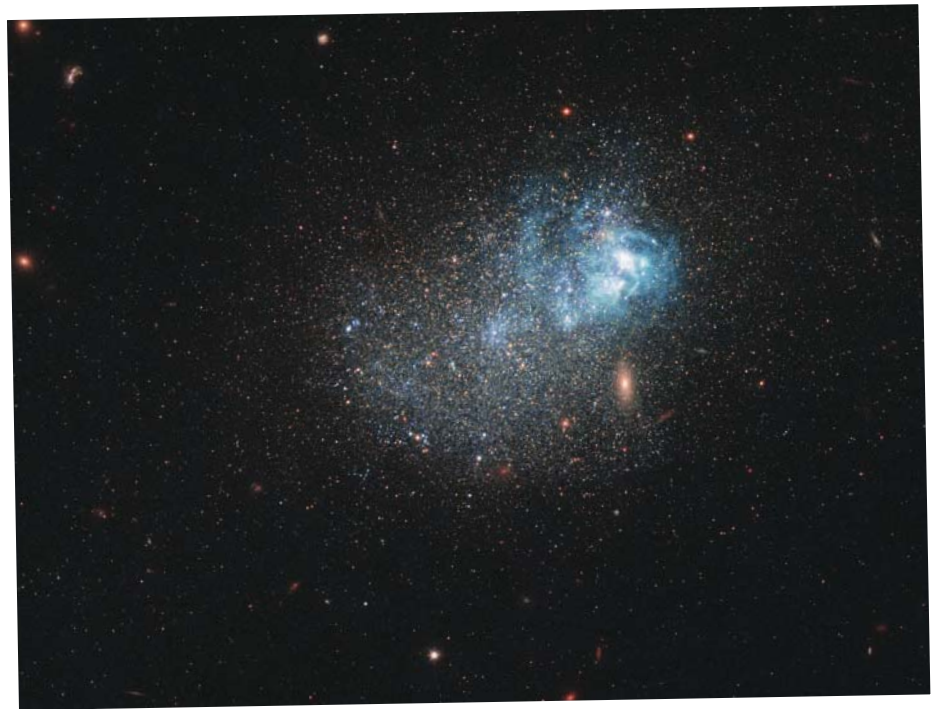
How can we use DNA to save populations at risk from habitat destruction, natural disasters, or low genetic diversity?



OPTION C

XNA

Could other life in the universe have evolved without DNA or RNA?



Assessment Criteria

Did I and my group...

- Develop one or more questions that provided opportunities for rich investigation? **QP**
- Develop effective methods to collect and record reliable data and information? **PC**
- Apply different ways of knowing to analyze, reflect on, and draw meaningful conclusions that are consistent with evidence? **PA**
- Consider and demonstrate an awareness of bias, and social, ethical, and environmental implications over the whole process of our inquiry? **E**
- Propose alternative course of thought and/or action that contribute to care for self, others, community, and world? **AI**
- Construct evidence-based argument using language, conventions, and representations appropriate for a specific purpose and audience? **C**